

EV Fleet Analytics & AI Platform

End-to-End Battery Health Monitoring, AI Peak-Shifting
& Zero-Touch RPA Data Pipeline

ENTERPRISE TECH STACK

- **Kaggle Telemetry Data** -> Raw CSV Dataset
- **Alteryx YXMD** -> ETL & Imputation Pipeline
- **Snowflake DW** -> Star Schema & CDC Streams
- **UiPath RPA** -> Orchestration & Auto-Refresh
- **Power BI & Fabric** -> 6-Page Dark Blue Dashboard
- **Streamlit AI Engine** -> Predictive SOH & Cost Model

14%

Cost Savings

SOH

Battery Health

0-Touch

RPA Pipeline

The EV Fleet Challenge

Fleet managers face 3 critical operational bottlenecks:

1. Battery Degradation Risks

Sudden roadside failures caused by unmonitored cell degradation below 80% State of Health (SOH).

2. Spiking Energy Utility Bills

Unoptimized peak-hour charging resulting in massive demand surcharges and high electricity costs.

3. Fragmented Telemetry Data

Telemetry scattered across charging stations, vehicle logs, and spreadsheets without unified analytics.

System Architecture

Modern automated data pipeline from ingestion to AI insights:

1

Telemetry Data Source

Kaggle EV Dataset (CSV Ingestion)

2

Alteryx YXMD Workflow

Null Imputation, Metric Calculation & Deduplication

3

Snowflake Data Warehouse

Raw/Staging/Core Star Schema, Streams & Tasks

4

UiPath RPA Orchestrator

Automated Snowflake SP Triggers & PBI Refresh

5

Power BI & Fabric Suite

6-Page Dark Obsidian Executive Dashboards

6

Streamlit AI Engine

SOH Degradation Model & Peak-Shift Cost Optimizer

Snowflake DW & Star Schema

High-performance enterprise data modeling in Snowflake:

Medallion Architecture

- RAW: Staging raw CSV streams
- STAGING: Cleaning & type casting
- CORE: Dimensional Star Schema (DIM_VEHICLE, FACT_CHARGING_SESSIONS)

CDC Streams & Tasks

- Automated Change Data Capture streams
- Scheduled tasks running daily transformations
- Zero manual pipeline intervention required

Stored Procedures

- SP_POPULATE_STAR_SCHEMA: Upsert core tables
- SP_CALCULATE_BATTERY_DEGRADATION: Identify high-risk battery cells below 80% SOH th

Zero-Touch RPA with UiPath

Automating repetitive data engineering tasks end-to-end:

1. Directory File Watcher

Monitors incoming telemetry CSV files from charging stations and uploads to Snowflake stage.

2. Stored Procedure Execution

Triggers Snowflake SPs to process raw data into Star Schema facts and dimensions.

3. Power BI Dataset Refresh

Sends REST API trigger to refresh Power BI & Fabric datasets automatically.

4. Executive Email Dispatch

Generates daily summary report PDF and dispatches anomaly alerts to fleet managers.

6-Page BI Dashboard Suite

Built with Microsoft Fabric Dark Obsidian design system:

1. Executive Summary

Fleet KPI cards, total energy used, cost growth & active vehicles.

2. Battery Health & SOH

Degradation trendlines, high-risk cell alerts (< 80% SOH).

3. Charging Efficiency

Peak vs Off-peak kWh usage, charger utilization rates.

4. Fleet Operations

Distance traveled, efficiency (kWh/mi), vehicle telemetry.

5. Energy Cost Analysis

Peak rate surcharges, monthly expenditure & savings breakdown.

6. AI Anomaly & Q&A

Natural language Q&A interface & ML degradation forecasts.

AI Engine & Cost Optimizer

Machine Learning & Smart Energy Scheduling:

14% Monthly Energy Cost Savings

Achieved via AI-driven peak-shifting charging algorithms.

Predictive Battery Degradation

Linear regression & ML model predicts SOH degradation trajectory and flags battery replacements before roadside failures occur.

Peak-Shifting Recommendations

Automatically shifts vehicle charging schedules from peak utility rate hours to off-peak night windows.

Natural Language Assistant

Streamlit-powered conversational AI engine enabling fleet managers to ask questions in plain English.

Enterprise Data Governance

Security, Compliance & Quality Assurance:

Role-Based Access Control (RBAC)

- Analyst Role: Read-only access to anonymized marts
- Manager Role: Full access to operational fleet data
- Executive Role: Aggregated KPI & cost view

Dynamic Data Masking & Encryption

- AES-256 encryption at rest and in transit
- Column-level masking on driver PII & vehicle IDs
- Full compliance with enterprise security standards

Data Quality & Audit Logging

- Automated quality exception matrix on nulls & bounds
- Complete line-item audit logs for all ETL runs
- Lineage tracking from source Kaggle CSV to Power BI

Key Business Impact & ROI

Quantifiable benefits delivered to fleet operations:

14%

Monthly Energy Cost Savings

Achieved by shifting charging schedules to off-peak rate windows.

< 80%

SOH Early Warning Threshold

Predictive alert system preventing unexpected battery cell failures.

100%

Zero-Touch Automation

UiPath RPA orchestrates ingestion, Snowflake SPs, and PBI refreshes.

THANK YOU FOR SWIPING!

Want to Explore the Code?

The full repository, SQL DDLs, DAX & Streamlit app are open-source!

Full GitHub Repository & Code:

Link available in the first comment below!

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- What data engineering stack do you use for IoT/EVs?
- Drop a comment or connect with me on LinkedIn!

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